

Toulouse INP - ENSEEIHT

SOUTENANCE DE STAGE DE FIN D'ÉTUDES

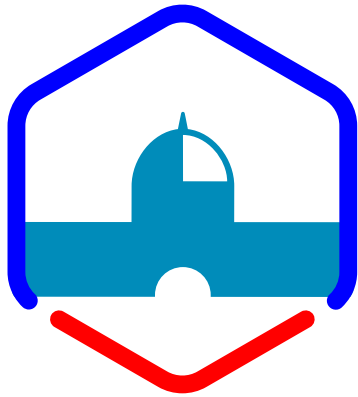
Ingénierie Logicielle Full-Stack et DevOps

Projet ONaCVG

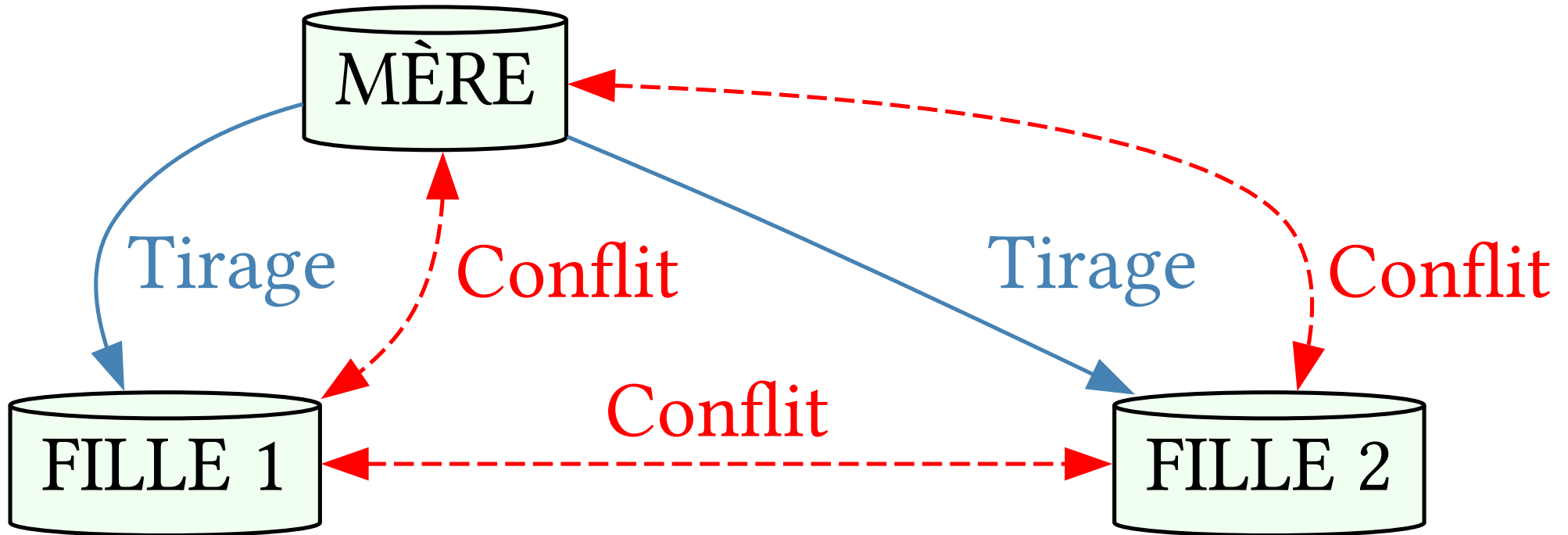
Vianney HERVY

16 mars 2026 - 18 septembre 2026

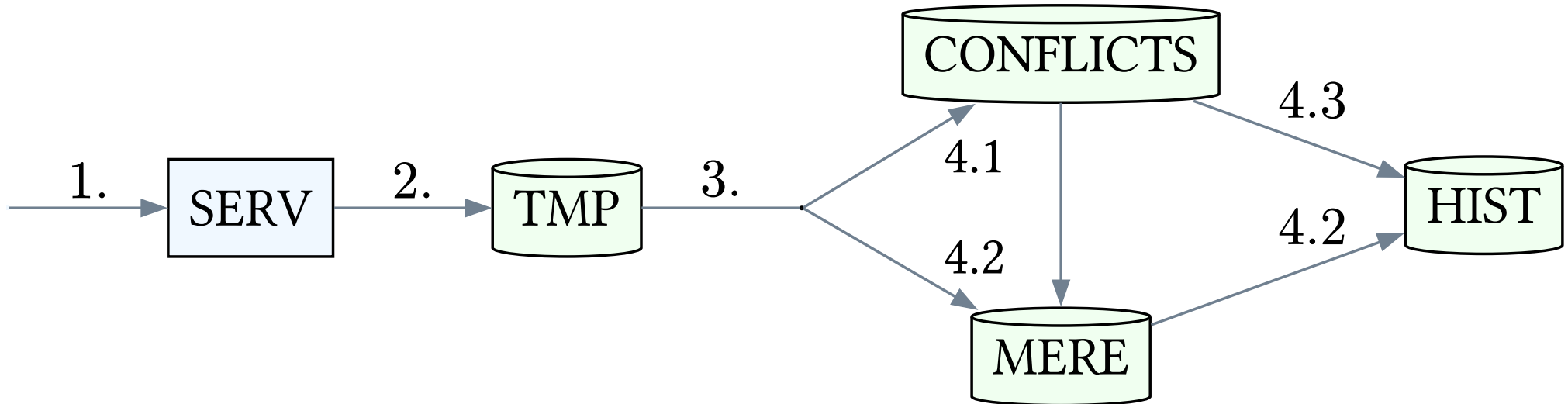
Le Contexte : l'ONaCVG



DÉFI 1 : Migration & Résolution de Conflits



DÉFI 1 : Architecture Asynchrone



DÉFI 1 : Les 3 Stratégies de Résolution

Détails du conflit 1863220

2 conflit(s) à résoudre

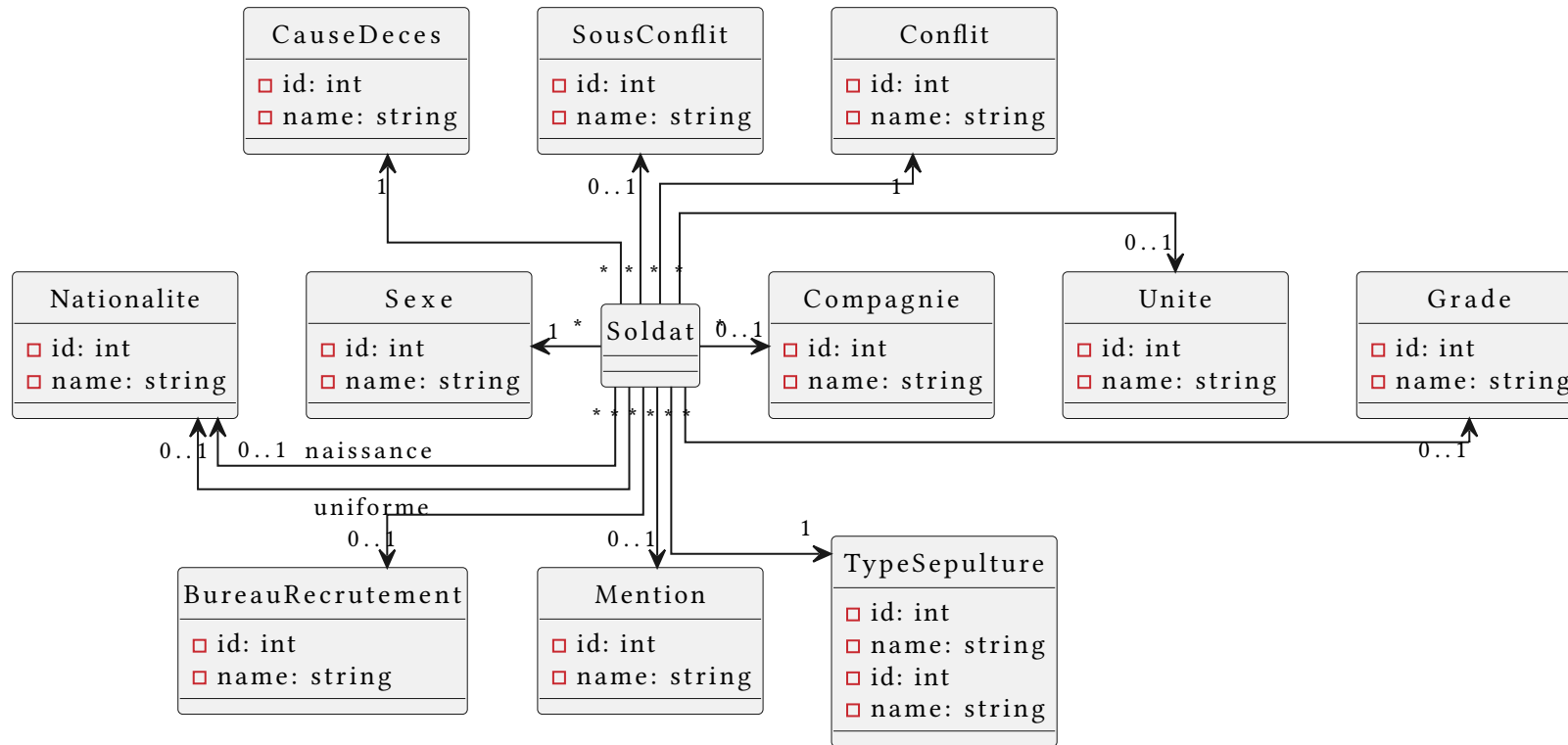
← Journal des conflits

NOM DU CHAMP	VALEUR MÈRE	Insérer avec un nouvel ID	Écraser l'entrée mère	Supprimer	Insérer avec un nouvel ID	Écraser l'entrée mère	Supprimer
orig_db	mere-export_b062_sepulture_B.csv	metz-sarrebourg-export_b062_sepulture_B.csv			raon-colmar-export_b062_sepulture_B.csv		
sdr_num	1863220	1863220			1863220		
sdr_num_site1	54467/ROZ	54467/ROZ			54467/ROZ		
sdr_num_conflit	3	3			3		
sdr_nom	BELON	BELON			BELON		
sdr_prenoms	Jean Baptiste Adrien	Jean Baptiste Adrien			Jean Baptiste Adrien		
sdr_grade	Sergent	Sergent			Sergent		
sdr_num_unite	134	134			134		
sdr_unite	R.I.	R.I.			R.I.		
sdr_classe	1910	1909			1909		
sdr_matricule	128	128			128		

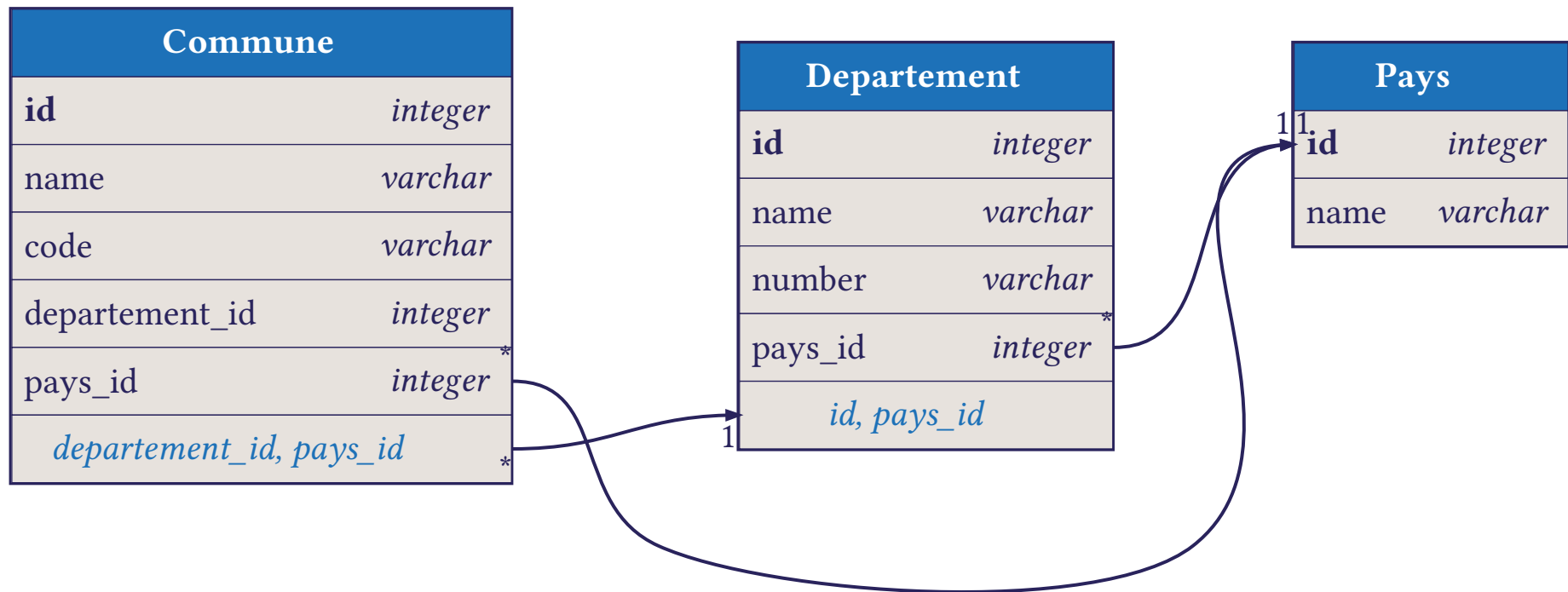
DÉFI 2 : Du plat au relationnel

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	[Saisie-Des-Registres].[N°]	N°Site1]	N°Conflit]	Nom]	Prenoms]	Grade]	N°Unite]	Unite]	Classe]	Matricule]	Recrutement]	Date-De-Naissance]	Lieu-De-Naissance]	Date-De-D	Memo-De	Lieu-de-D	Date-1ere	Lieu-1ere	Date-Dern	Mention-D	Droit-SP]	N°Nator
2	1447929	"55012/CAA"	"3"	"BARETH"	"Paul Alexandre"	"Sergent"	"7"	"Gr. Cyclistes"	"1913"	"470"	"Nancy (54 - Meurthe-et-Moselle)"	"28/02/1893"	"Bürglen (Suisse)"	"27/09/1911"	"	"Loupmor"	"	"	"Vrai"	"Vrai"	"100000"	
3	1448118	"55012/MAR"	"3"	"BERTRAND"	"Rolland"	"	"6"	"R.A."	"1911"	"	"	"	"	"	"	"Apremor"	"	"Cimetière"	"04/04/1911"	"Vrai"	"Vrai"	"100000"
4	1448206	"55012/MAR"	"3"	"BOUYOUX"	"Auguste"	"	"85"	"R.I."	"1904"	"	"	"	"	"	"	"Apremor"	"	"Marbott"	"05/04/1911"	"Vrai"	"Vrai"	"100000"
5	1449138	"55012/MAR"	"3"	"BIDAUD"	"François Joseph"	"Sergent"	"10"	"R.I."	"1911"	"306"	"Mâcon (71 - Saône-et-Loire)"	"30/04/1891"	"Chauffailles (71 - Saône-et-Loire)"	"26/06/1911"	"	"Han-sur-Meuse"	"	"Cimetière"	"25/04/1911"	"Vrai"	"Vrai"	"100000"
6	1449718	"55012/MAR"	"3"	"BOUGUIN"	"René"	"Aspirant"	"85"	"R.I."	"1914"	"	"	"	"	"09/06/1911"	"	"Apremor"	"	"Cimetière"	"11/05/1911"	"Vrai"	"Vrai"	"100000"
7	1449809	"55012/MAR"	"3"	"BONNETETE"	"Louis"	"	"56"	"R.I."	"1911"	"	"	"	"	"23/05/1911"	"	"Apremor"	"	"Cimetière"	"12/05/1911"	"Vrai"	"Vrai"	"100000"
8	1449894	"55012/MAR"	"3"	"BASTIAN"	"Henri"	"	"171"	"R.I."	"1901"	"	"	"	"	"14/01/1911"	"	"Apremor"	"	"Cimetière"	"12/05/1911"	"Vrai"	"Vrai"	"100000"
9	1449983	"55012/MAR"	"3"	"BIGOT"	"Gaston"	"	"25"	"B.C.P."	"1911"	"	"Reims (51 - Marne)"	"	"	"17/11/1911"	"	"Bois D'ai"	"	"	"	"Vrai"	"Vrai"	"100000"
10	1449998	"55012/MAR"	"3"	"BEAUJARD"	"Georges Albert"	"	"10"	"R.I."	"1915"	"7.705"	"Bourges (18 - Cher)"	"	"	"07/07/1911"	"	"Bois D'ai"	"	"	"	"Vrai"	"Vrai"	"100000"
11	1450030	"55012/MAR"	"3"	"BRIDAY"	"Marius"	"	"29"	"R.I."	"1914"	"876"	"Mâcon (71 - Saône-et-Loire)"	"	"	"	"	"Bois De l'	"	"La Louvière"	"11/05/1911"	"Vrai"	"Vrai"	"100000"
12	1450048	"55012/MAR"	"3"	"BONNAUD"	"	"	"	"	"	"	"	"	"	"	"	"Han-sur-Meuse"	"	"Bois D'ai"	"02/07/1911"	"Vrai"	"Vrai"	"100000"
13	1450591	"55023/AVO"	"3"	"BREGERE"	"Armand Pierre"	"	"161"	"R.I."	"1906"	"2506"	"Rennes (35 - Ille-et-Vilaine)"	"	"	"21/04/1911"	"	"	"	"	"20/11/1911"	"Vrai"	"Vrai"	"100000"
14	1450712	"55023/AVO"	"3"	"BONNARD"	"Oscar Joseph"	"	"135"	"R.I."	"1905"	"19427"	"Abbeville (80 - Somme)"	"	"	"07/05/1911"	"	"	"	"Bois Carré"	"06/05/1911"	"Vrai"	"Vrai"	"100000"
15	1451113	"55023/AVO"	"3"	"BOUCHE"	"Léon"	"	"	"	"	"	"	"	"	"	"	"	"	"Route D'	"30/10/1911"	"Vrai"	"Vrai"	"100000"
16	1451307	"55023/AVO"	"3"	"BERGER"	"Jean Baptiste"	"Caporal"	"290"	"R.I."	"1904"	"1640"	"Châteauroux (36 - Indre)"	"	"	"04/05/1911"	"	"	"	"Cote 304"	"30/01/1911"	"Vrai"	"Vrai"	"100000"
17	1451678	"55023/AVO"	"3"	"BRISSON"	"Pierre"	"	"125"	"R.I."	"1901"	"1116"	"Poitiers (86 - Vienne)"	"	"	"07/05/1911"	"	"	"	"Sud Cote"	"04/10/1911"	"Vrai"	"Vrai"	"100000"
18	1451719	"55023/AVO"	"3"	"BRENE"	"	"	"	"	"	"	"	"	"	"	"	"	"	"Prairie D'	"30/06/1911"	"Vrai"	"Vrai"	"100000"
19	1451836	"55023/AVO"	"3"	"BOUTELOUP"	"Louis Nicolas"	"	"92"	"R.I."	"1914"	"191"	"Tulle (19 - Corrèze)"	"	"	"08/03/1911"	"	"Bois des"	"	"Bois Des"	"	"Vrai"	"Vrai"	"100000"
20	1451913	"55023/AVO"	"3"	"BENOIT"	"Daniel Théophile"	"	"153"	"R.I."	"1908"	"231"	"Pont-Saint-Esprit (30 - Gard)"	"	"	"07/04/1911"	"	"	"	"Haucourt"	"15/05/1911"	"Vrai"	"Vrai"	"100000"
21	1452176	"55023/AVO"	"3"	"BUGEANT"	"Zéphyr Octave"	"	"155"	"R.I."	"1914"	"1426"	"Châtelleraut (86 - Vienne)"	"	"	"29/05/1911"	"	"	"	"Mort Ho"	"24/02/1911"	"Vrai"	"Vrai"	"100000"
22	1452288	"55023/AVO"	"3"	"BROCHARD"	"Benoni Eugène F."	"2°Classe"	"287"	"R.I."	"1913"	"1069"	"La Roche-sur-Yon (85 - Vendée)"	"27/03/1893"	"Les Sables-d'Oloron"	"20/05/1911"	"	"Mort Ho"	"	"Mort Ho"	"25/05/1911"	"Vrai"	"Vrai"	"100000"
23	1456840	"55051/CAL"	"4"	"BROOKE"	"Kenneth Arba"	"Sgt."	"103"	"R.A.F. (V.R.)"	"	"1486403"	"	"	"	"15/07/1911"	"	"	"	"	"	"Vrai"	"Vrai"	"99132"
24	1456846	"55053/CAL"	"3"	"BERGE"	"Julien Eugène"	"2°Classe"	"20"	"R. de Chasset"	"1911"	"3734"	"Blois (41 - Loir-et-Cher)"	"12/01/1891"	"Ménestreau-en-Touraine"	"13/08/1911"	"	"Entre Pie"	"	"	"	"Vrai"	"Vrai"	"100000"
25	1459308	"55073/BSM"	"3"	"BEUCHER"	"Henri Joseph"	"	"39"	"R.I."	"1913"	"585"	"Laval (53 - Mayenne)"	"	"	"23/06/1911"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
26	1461069	"55073/BSM"	"3"	"BALMON"	"Martial"	"Caporal"	"350"	"R.I."	"1897"	"509"	"Bordeaux (33 - Gironde)"	"	"	"23/05/1911"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
27	1461298	"55073/BSM"	"3"	"BAUMONT"	"Camille"	"	"221"	"R.I."	"1903"	"673"	"Langres (52 - Haute-Marne)"	"	"	"07/07/1911"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
28	1461311	"55073/BSM"	"3"	"BROCHEC"	"Charles Marie"	"Caporal"	"225"	"R.I."	"1910"	"3147"	"Brest (29 - Finistère)"	"	"	"09/11/1911"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
29	1461321	"55073/BSM"	"3"	"BOURIQUET"	"François Ferdinand"	"	"416"	"R.I."	"1905"	"1476"	"Guéret (23 - Creuse)"	"	"	"08/01/1911"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
30	1461368	"55073/BSM"	"3"	"BLOY"	"Jean"	"	"24"	"B.C.A."	"1911"	"289"	"Foix (09 - Ariège)"	"	"	"29/10/1911"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
31	1461503	"55073/BSM"	"3"	"BODSON"	"René Hippolyte"	"Caporal"	"165"	"R.I."	"1906"	"684"	"Mézières (08 - Ardennes)"	"	"	"14/12/1911"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
32	1461537	"55073/BSM"	"3"	"BRIERE"	"Cyprien Armand"	"	"2"	"R.T.A."	"1906"	"2428"	"Rouen Nord (76 - Seine-Maritime)"	"	"	"25/11/1911"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
33	1461578	"55073/BSM"	"3"	"BAUD"	"Pierre Jean Marie"	"	"11"	"R.I."	"1912"	"422"	"Brest (29 - Finistère)"	"	"	"03/08/1911"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
34	1461685	"55073/BSM"	"3"	"BROUILLET"	"Louis Auguste"	"	"67"	"R.I."	"1902"	"313"	"Soissons (02 - Aisne)"	"	"	"24/09/1911"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
35	1462213	"55078/BLM"	"3"	"BERTIAUX"	"Alfred Eugène"	"	"31"	"R.I."	"1901"	"545"	"Seine 5°B (75 - Paris (e))"	"15/09/1881"	"Gisors (27 - Eure)"	"13/12/1911"	"	"Eclisfont"	"	"Eclisfont"	"03/02/1911"	"Vrai"	"Vrai"	"100000"
36	1463959	"55106/CTT"	"3"	"BARDS"	"Marcel"	"	"70"	"R.I."	"	"	"	"	"	"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
37	1464025	"55106/CTT"	"3"	"BATONNIER"	"Louis Edouard"	"	"155"	"R.I."	"1912"	"	"Châlons-sur-Marne (51 - Marne)"	"	"	"08/04/1911"	"	"	"	"Ferme D'	"20/04/1911"	"Vrai"	"Vrai"	"100000"
38	1464490	"55106/CTT"	"3"	"BARELLA"	"Joseph"	"	"	"	"1895"	"800"	"Narbonne (11 - Aude)"	"	"	"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
39	1464723	"55106/CTT"	"3"	"BELLIARD"	"Victor Jean Augu"	"	"306"	"R.I."	"1914"	"	"	"	"	"30/05/1911"	"	"	"	"Chattanc"	"19/08/1911"	"Vrai"	"Vrai"	"100000"
40	1464754	"55106/CTT"	"3"	"BENARD"	"Victor"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"Vrai"	"Vrai"	"100000"
41	1465477	"55106/CTT"	"3"	"BRURIAUX"	"Edmond"	"Caporal"	"254"	"R.I."	"1893"	"1462"	"Seine 5°B (75 - Paris (e))"	"	"	"23/05/1911"	"	"	"	"Cumière"	"	"Vrai"	"Vrai"	"100000"

DÉFI 2 : Modélisation en étoile (Thésaurus)



DÉFI 2 : La contrainte astucieuse



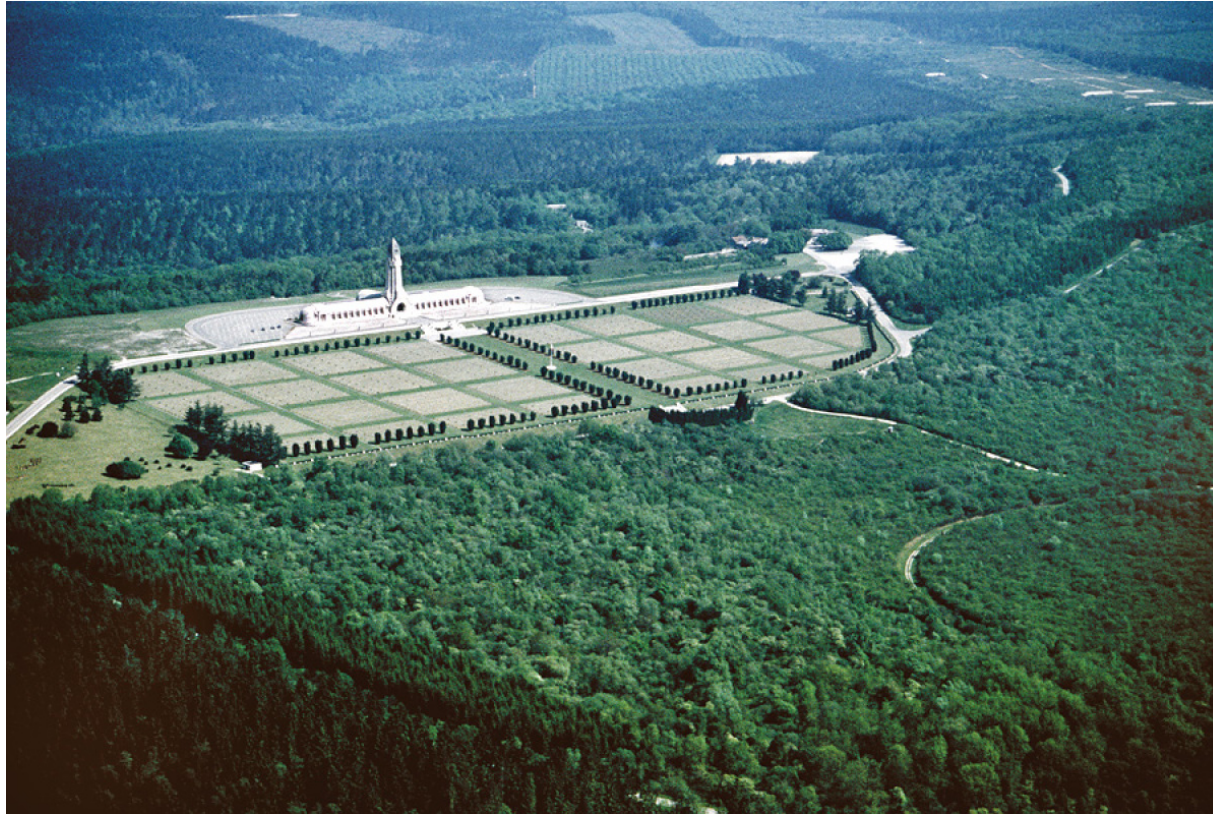
DÉFI 3 : Thésaurus & Normalisation

Mémoire *des*

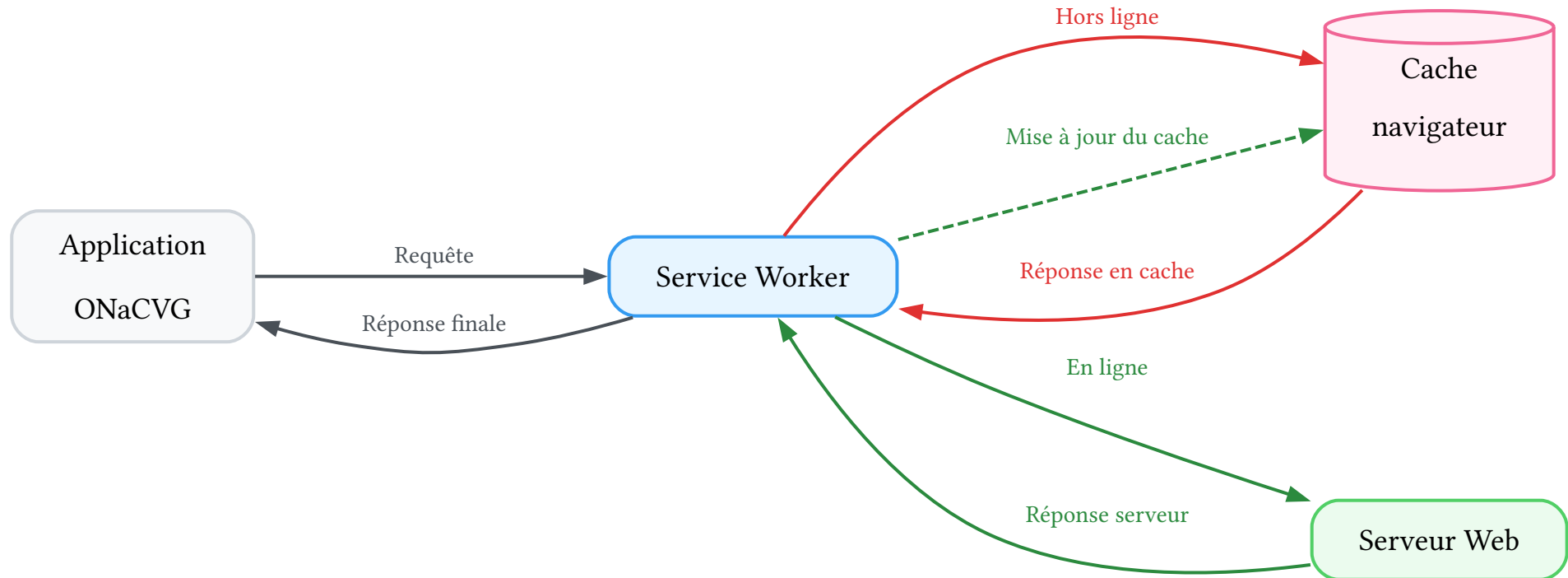
HOMMES

PORTAIL CULTUREL DU MINISTÈRE DES ARMÉES

DÉFI 4 : Inspection hors-ligne



DÉFI 4 : Le cache du navigateur



Conclusion

Questions ?

Merci de votre attention.

Bonus

- **Philosophie** : « Make Illegal States Unrepresentable » (leçon tirée d'OCaml et Ada, appliquée à PHP).
- **Pragmatisme** : La qualité et la rigueur d'un logiciel ne viennent pas que du langage, mais de ses outils (PHPStan niveau 9, CI stricte).
- **Impact** : Apporter une vraie solution de terrain pour des agents de l'ONaCVG, donnant du sens au code.